Toward Conjecture 12

A best-effort attack: partial results, a barrier theorem, and a proof plan :: August 29, 2026

This note does two things. Section 1 records my assessment of the summary note for Fede. The rest is the honest state of my attack on Conjecture 12 ($|\mathcal{C}| \geq 2$ implies every nice test is admissible): a master lemma that unifies every admissibility proof we have (Section 2); a barrier theorem showing that this method *provably cannot* decide the remaining cases (Section 3); two rigidity results that cut off the natural routes to a counterexample (Sections 4 and 5); a reduction of the two-bad-facet case to a single scalar inequality (Section 6); and the proof plan that all of this points to (Section 7). I could not prove the conjecture. I believe the material below is the reason no one-step proof exists, and it isolates the one missing lemma. Compact proofs of the results stated without proof in the summary note are collected in the appendix.

1 Assessment of the summary note

The note is accurate: I verified every stated result against the arguments we have. Suggested repairs, all minor: the result numbering is nonsequential (Lemma 1, Lemma 6, Definition 3, ...) and should be renumbered; "or move involved" should read "or more involved"; in Definition 3 the polyhedral criterion should note that irredundancy makes $b_j = h_A(a_j)$, so the halfspaces are supporting; and the claim that the redundant-only test is inadmissible for $\mathcal{C} = \{1\}$ in every dimension deserves its one-line justification — the test depends only on (X_1, X_r), the projections of Λ_0 and Λ_1 fill the bivariate null and alternative, and the bivariate dominator therefore lifts (its level and power depend only on the (X_1, X_r)-marginal, which is bivariate normal with correlation Ω_{1r}, covered by the theorem for every ρ). Two additions worth making: a sentence scoping the bivariate necessity via $\Pi_\rho = \Omega_\rho \Lambda_1^*$, so the reader sees why necessity is a $\Lambda_1^* \neq \{0\}$ phenomenon; and a description of the open frontier — regions with at least two essential infeasible facets, with the canonical minimal examples (a rectangle with $|\mathcal{R}| \geq 2$; a rectangle plus one binding infeasible cut; the tilted-gate family).

2 The constraint set and the separating-score method

Fix $|\mathcal{C}| \geq 2$ and a nice A, and let ψ be a level-α test that weakly dominates ϕ_A; put $h := \psi - \phi_A$ and $M := \mathrm{E}_0 h^+ = \mathrm{E}_0 h^-$. Write $\widetilde{A}$ for the *feasible hull* of A: the intersection of its supporting halfspaces with normals in $\mathcal{F}(\Omega, \Lambda)$.

Lemma 2.1 (the constraint set)**.** *(i)* $\{h > 0\} \subseteq A$ *and* $\{h < 0\} \subseteq A^c$*; (ii)* $\mathrm{E}_0 h = 0$*; (iii)* $q := \mathrm{E}_0[Xh] = 0$*; (iv)* $\mathrm{E}_0[e^{\theta' X} h] \geq 0$ *for every* $\theta \in \Omega^{-1}\Lambda_1$*; (v)* $h^- = 0$ *a.s. on* $\widetilde{A}^c$*, so* h^- *is supported on* $\widetilde{A} \setminus A$*.*

The proofs are those of the earlier notes (appendix). Every admissibility result we have follows from Lemma 2.1 through one scalar comparison:

Lemma 2.2 (separating score)**.** *Suppose there exists*

$$k(x) = c_0 + \ell' x + \sum_{i=1}^{m} \nu_i \, e^{\theta_i' x}, \qquad \nu_i \geq 0, \quad \theta_i \in \Omega^{-1}\Lambda_1, \tag{1}$$

with $k \leq 0$ on A and $k > 0$ a.e. on $\widetilde{A} \setminus A$. Then ϕ_A is α-admissible.

Proof. By (ii)–(iv), $\mathrm{E}_0[kh] = c_0 \mathrm{E}_0 h + \ell' q + \sum_i \nu_i \mathrm{E}_0[e^{\theta_i' X} h] \geq 0$. By (i) and (v), $\mathrm{E}_0[kh] = \int_A kh^+ - \int_{\widetilde{A}\setminus A} kh^- \leq -\int kh^- < 0$ whenever $M > 0$. Hence $M = 0$, $h = 0$ P_0-a.s., and mutual absolute continuity finishes the argument. □

Feasible support is the case $\widetilde{A} = A$ (no score needed); one essential bad facet is the case $k = u'x - \kappa$, linear. The natural question is whether Lemma 2.2 can reach the frontier regions. It cannot.

3 A barrier theorem

Take the canonical frontier case: $\Omega = I_4$, $\mathcal{C} = \{1, 2\}$, $\mathcal{R} = \{3, 4\}$, and the exact-size rectangle $A = \bigcap_{j=1}^{4} \{x_j \leq \kappa_j\}$, whose two bad facets are the redundant faces e_3, e_4.

Theorem 3.1 (no separating score exists)**.** *For this A, no function k of the form* (1) *satisfies $k \leq 0$ on A and $k > 0$ on $\widetilde{A} \setminus A$.*

Proof. First, each θ_i must be nonnegative: if $\theta_{ij} < 0$, move from any $y^0 \in A$ along $-e_j$ (which stays in A); the ith term grows exponentially while $|\ell' x|$ grows linearly and the other terms are nonnegative, so $k > 0$ somewhere on A. Being also in Λ_1, each θ_i is feasible, and any nonzero feasible $\theta \geq 0$ has $\theta_1 + \theta_2 > 0$ (if $\theta_1 = \theta_2 = 0$, feasibility forces $\theta_3 = \theta_4 = 0$). Second, $k \leq 0$ on A as $x_j \to -\infty$ forces $\ell \geq 0$.

Now consider the gap points $z_T = (-T, -T, \kappa_3 + 1, \kappa_4 - 1)$. They violate the third face, satisfy the others, and lie in $\widetilde{A}$ for large T: for feasible θ, $\theta' z_T - \theta'\kappa = \theta_3 - \theta_4(\cdot) - (T + \kappa_1)\theta_1 - (T + \kappa_2)\theta_2 \leq (\theta_1 + \theta_2)(1 - T - \min_j \kappa_j) \leq 0$ using $\theta_3 \leq \max(\theta_1, \theta_2) \leq \theta_1 + \theta_2$. Since every θ_i has $\theta_{i1} + \theta_{i2} > 0$, all exponential terms vanish as $T \to \infty$, so $k(z_T) = c_0 + \ell' z_T + o(1)$. Positivity for all T forces $\ell_1 + \ell_2 \leq 0$, hence $\ell_1 = \ell_2 = 0$; the variant with $x_4 \to -\infty$ (still in the gap and in $\widetilde{A}$) forces $\ell_4 = 0$, and the symmetric fourth-face lobe forces $\ell_3 = 0$. So $\ell = 0$ and $k = c_0 + \sum_i \nu_i e^{\theta_i' x}$. If $c_0 > 0$, then $k(z_T) \to c_0$ is fine but deep points of A give $k \to c_0 > 0$, violating $k \leq 0$ on A. If $c_0 < 0$, then $k(z_T) \to c_0 < 0$, violating gap positivity. If $c_0 = 0$, then $k \geq 0$ everywhere, and $k \leq 0$ on A forces $\nu = 0$, i.e. $k \equiv 0$, violating gap positivity. Contradiction in every case. □

The same argument runs for the three-dimensional tilted-gate family $A(\lambda) = \{x_1 \leq \kappa, x_2 \leq \kappa, x_1 + \lambda x_3 \leq b, x_2 + \lambda x_3 \leq b\}$, $\lambda > 1$: each gate's "dead lobe" (both cores escaping to $-\infty$ along the gate) forces ℓ proportional to that gate's normal, and the two lobes together force $\ell = 0$, after which the constant-term trichotomy applies.

The interpretation is sharp. Every admissibility proof we possess aggregates Lemma 2.1 into one scalar comparison, and Theorem 3.1 says no such aggregation can decide the frontier. The obstruction lives in the *dead zones*: gap regions escaping to $-\infty$ in all core coordinates, where every alternative tilt vanishes and only the (sign-constrained) linear terms reach — and two bad facets make incompatible demands on the single linear term. A proof of Conjecture 12 therefore needs either a new confinement principle that empties the dead zones before scoring, or a genuinely non-scalar use of the constraints; and a counterexample, if one exists, must place comparison mass in the dead zones. The next two sections squeeze the second possibility.

4 No dominator hides in a dead slab

Proposition 4.1 (slab rigidity)**.** *Let $\Omega = I_d$ and let A be the exact-size rectangle (any $|\mathcal{C}| \geq 1$, any $|\mathcal{R}|$). Fix $T > 0$ and suppose $h = 1\{x_1 \leq -T\}\, g(x_2, \ldots, x_d)$ satisfies the conditions of a dominance perturbation (Lemma 2.1(i)–(ii) and $\mathrm{E}_\mu h \geq 0$ for all $\mu \in \Lambda_1$). Then $g = 0$ a.s.*

Proof. By independence, $\mathrm{E}_\mu h = \Phi(-T - \mu_1)\, \mathrm{E}_\nu g$ with $\nu = (\mu_2, \ldots, \mu_d)$. As $\mu_1 \to \infty$ the constraint $\mu_r \leq \max_c \mu_c$ frees ν completely, so $\mathrm{E}_\nu g \geq 0$ for *every* $\nu \in \mathbf{R}^{d-1}$, and (ii) gives $\mathrm{E}_0 g = 0$. The sign structure makes g a dominance perturbation of the $(d-1)$-dimensional section rectangle against the *unrestricted* alternative. In the unrestricted problem every direction is an alternative direction, so every closed convex acceptance region is feasibly supported, and the confinement argument of Lemma 2.1(v) applies to every supporting halfspace of the section: $g^- = 0$ a.s., and $\mathrm{E}_0 g = 0$ then gives $g = 0$ a.s. □

So the dead zones, where the barrier lives, cannot host a self-contained dominator: a slab of a $|\mathcal{C}| \geq 2$ problem faces an effectively unrestricted alternative, which is the Birnbaum regime where convexity already wins. Any dominator must couple dead zones to live zones.

5 The natural coupling fails: localized gates

There is exactly one natural family of couplings: borrow the $|\mathcal{C}| = 1$ gated dominator and arm it only when the data indicate the other core is dead. In the three-dimensional two-core model with the equal-cutoff rectangle, let g_{cr} denote the calibrated gate swap in coordinates (c, r) (reject the diagonal corner G_{cr}, accept the strip D_{cr}, with $P_0(G_{cr}) = P_0(D_{cr})$), and define

$$h_L := 1\{x_1 \leq -L\}\, g_{23}(x_2, x_3) + 1\{x_2 \leq -L\}\, g_{13}(x_1, x_3)\ .$$

This satisfies the sign structure and $\mathrm{E}_0 h_L = 0$, and it gains where it was designed to: numerically, at $\mu = (2, -5, 1)$ (core 1 active, core 2 dead), $\mathrm{E}_\mu h_L \simeq +0.108$. But it violates the first-moment identity: $q_2 = \Phi(-L)\, \mathrm{E}_0[X_2 g_{23}] > 0$, and Lemma 2.1(iii) says a dominator must have $q_2 = 0$. The violation is realized at near-null alternatives with a dead core: at $\mu = (t, -3, 0) \in \Lambda_1$ the armed $(2,3)$-gate loses a fixed amount, $\Phi(-L)\Delta_{23}(-3, 0) < 0$, while the compensating $(1,3)$-gain vanishes linearly in t. Numerically with $L = 2$, $\alpha = 0.05$:

$\mathrm{E}_\mu h_L \simeq +0.0113$ at $t = 0.5$, $+0.0017$ at $t = 0.1$, and -2.8×10^{-4} at $t = 0.005$. Repairing this requires a switch with $\mathrm{E}_0[\text{switch}] = 0$, which the sign structure forbids (the negative part of the switch times the negative part of the gate puts $h > 0$ outside A). I regard this as the strongest concrete evidence for the conjecture: the only known source of power improvements, transplanted in its most favorable form, is killed by the first-moment identity exactly where that identity has bite.

6 The two-facet case is one scalar inequality

Lemma 6.1 (absolute-value reduction)**.** *Let $A = \widetilde{A} \cap \{g_u \le 0\} \cap \{g_w \le 0\}$ be nice with two essential bad facets, where $g_u(x) = u'x - \kappa_u$, $g_w(x) = w'x - \kappa_w$. If $M > 0$, then*

$$\mathrm{E}_0\big[\,|g_u(X) - g_w(X)|\,h\big] < 0\ .$$

Consequently, Conjecture 12 holds for all such A if and only if every dominance perturbation satisfies $\mathrm{E}_0[\,|(u-w)'X - (\kappa_u - \kappa_w)|\,h] \ge 0$.

Proof. Let $f = \max(g_u, g_w)$. Then $f \le 0$ on A and $f > 0$ a.e. on $\widetilde{A} \setminus A$, so as in Lemma 2.2, $M > 0$ implies $\mathrm{E}_0[fh] < 0$. But $f = \frac{1}{2}(g_u + g_w) + \frac{1}{2}|g_u - g_w|$, and the affine part integrates to zero against h by (ii)–(iii). □

For $A(\lambda)$ the relevant direction is $u - w \propto (1, -1, 0)$, and *both* $\pm(1, -1, 0)$ lie in Λ_1, so two-sided exponential information $\mathrm{E}_0[\cosh\{t(X_1 - X_2)\}h] \ge 0$ is available for every t. The missing step is exactly that $|z|$ is not a mixture of $\cosh(tz)$'s and affine functions — every such mixture grows superlinearly — which is Theorem 3.1 again in scalar form. For the four-dimensional rectangle, $u - w = e_3 - e_4$ and neither sign is in Λ_1, so even this reduction has less to work with; the rectangle with $|\mathcal{R}| \ge 2$ is, I believe, the hardest case.

7 The proof plan

The results above locate the single missing ingredient.

Step 1 (the key lemma: dead-zone confinement). Prove that every dominance perturbation has $h = 0$ a.s. on the dead zones — gap regions along which all coordinates in some core-detecting set escape to $-\infty$. Proposition 4.1 proves this for product-form h; the open problem is the coupled case. The natural tool is a conditional argument in the spirit of the bivariate rearrangement lemma: condition on the deep core coordinates and use the fact that the conditional problem is effectively unrestricted, where the section of a nice region is convex and hence rigid. The technical difficulty is that dominance constrains only the x_1-integral of the conditional perturbations, not each slice.

Step 2 (score on the truncated gap). Granted Step 1, h^- is confined to the part of $\widetilde{A} \setminus A$ where some feasible tilt stays active. There the barrier computation no longer applies, and finite mixtures (1) plausibly separate: each lobe of the truncated gap is exposed by tilts near the feasible boundary of its gate normal. This should be checked first numerically (a small

convex program: minimize the gap infimum of k minus the A-supremum over (c_0, ℓ, ν, θ)), then proved for the canonical regions.

Step 3 (the decisive experiment, in parallel). The discretized linear program from the earlier note, aimed at $A(2)$ and at the four-dimensional rectangle, with the $1 \leftrightarrow 2$ symmetry imposed and the alternative net including deep rays. A robustly positive value yields the counterexample; a vanishing value yields, through the dual solution, the weights of the separating construction that Step 2 needs — including any capacity multipliers (dual variables on $0 \le \psi \le 1$), which are exactly the terms the score class (1) lacks and which a Blyth-type argument would build implicitly.

Where I stand. Sections 4 and 5 close the two natural routes to a counterexample; Section 3 explains why no proof has appeared. My credence in Conjecture 12 has gone up: the dead zones are the only place a dominator could live, and they are inert both in isolation (Proposition 4.1) and under the one natural coupling (Section 5). But the conjecture is genuinely open, and Step 1 is where I would spend effort — it is a self-contained lemma, its product-form case is already proved, and it converts the barrier from an obstruction into a two-line finish.

A Proofs of the results stated in the summary note

Lemma 1 (dual-cone dichotomy). For $\mathcal{C} = \{1\}$: normalizing $\mu_1 = 1$, $v_1 + \sum_r \mu_r v_r \ge 0$ for all $\mu_r \le 1$ forces $v_r \le 0$ (let $\mu_r \to -\infty$) and then $v_1 + \sum_r v_r \ge 0$ (at $\mu_r = 1$); conversely these suffice, and the cone representation is a direct check. For $|\mathcal{C}| \ge 2$: $e_c + s e_{c'} \in \Lambda_1$ for all real s kills the core coordinates pairwise; means with one redundant coordinate at the core maximum and the others at $-M$ kill the redundant coordinates.

Lemma 6 (first-order restriction). $\mu = 0 \in \Lambda_0$ gives $\mathrm{E}_0\psi \le \alpha = \mathrm{E}_0\phi$; dominance along $te_c \in \Lambda_1$ and continuity of Gaussian power functions give the reverse, so $\mathrm{E}_0 h = 0$. For $\mu \in \Lambda_1$, $t \mapsto \mathrm{E}_0[h e^{t(\Omega^{-1}\mu)'X}]$ is nonnegative for $t > 0$ (dominance along $t\mu$, a ray in the cone Λ_1), vanishes at $t = 0$, and is differentiable there ($|h| \le 1$, Gaussian m.g.f.), so its right derivative $(\Omega^{-1}\mu)'q$ is nonnegative; hence $\Omega^{-1}q \in \Lambda_1^*$.

Proposition 4 (feasible support suffices). For nonzero $a \in \mathcal{F}(\Omega, \Lambda)$ with $h_A(a) < \infty$: $\Omega a \in \Lambda$ and $a'\Omega a > 0$ put $\Omega a \in \Lambda_1$, so $\mathrm{E}_0[e^{ta'X}h] \ge 0$ for all $t > 0$; since h^+ lives on $A \subseteq \{a'x \le h_A(a)\}$, letting $t \to \infty$ kills h^- above the supporting plane. Lindelöf reduces the union of the half-space complements to a countable one, so $h^- = 0$ a.s. on A^c when A is feasibly supported; $\mathrm{E}_0 h = 0$ then gives $h = 0$ a.s.

Proposition 5. If all finite normals are feasible, the feasible supporting halfspaces include all supporting halfspaces, whose intersection is A. Conversely with $|\mathcal{C}| = 1$, $\mathcal{F}$ is a closed convex cone (the defining constraint is linear), the barrier cone of A lies in the closed convex cone generated by the feasible normals used in the representation, hence in $\mathcal{F}$.

Proposition 7 (one arbitrary halfspace). Confinement puts h^- on $\{u'x > \kappa\} \cap A_f$; h^+ lives on $A \subseteq \{u'x \le \kappa\}$; if $M > 0$ then $\mathrm{E}_0[(u'X - \kappa)h] < 0$, contradicting $u'q = 0$ from Lemma 6 and the dichotomy.

Corollaries 8–9 and core-only rectangles. Corollary 8: $A_f = \mathbf{R}^d$, $u = a$. Corollary 9: core faces are feasible since $(\Omega e_c)_r = \Omega_{rc} < 1 = (\Omega e_c)_c$; the single redundant face is the extra halfspace. Core-only rectangles: all facets feasible by the same computation, so Proposition 4 applies for any $|\mathcal{C}| \geq 1$.